Name: Sri hari siddharth M
Roll No: 241201055
Year & Dept: 3 & MCT

Experiment 2: Preparation of a Software Project Plan for a Sample Application

1. Aim

To prepare a software project plan covering scope, timeline and team roles for a sample application, a web-based Library Management System for a college.

2. Background and Objectives

A college library currently maintains membership records, book issue registers and fine calculations on paper and in separate spreadsheets. Staff spend significant time searching for books, students cannot check availability remotely, and management lacks reliable circulation statistics. A web-based system is proposed to bring all these functions into one application.

The objectives of the project are: to reduce manual register maintenance, to provide real-time availability of books, to automate fine calculation for overdue returns, to give the librarian reports on circulation and popular titles, and to deliver the system within a fixed schedule of about eleven weeks.

3. Project Scope

- In scope: student and staff membership registration, book catalogue management with search, issue and return of books, automatic fine calculation, reminder notifications, and a reports dashboard.
- Out of scope: hosting of e-books, inter-library loan network, native mobile applications, and online payment gateway integration.
- Assumptions: a single campus server, existing book data available in spreadsheet form, a stable college network, and librarian staff available for acceptance testing.
- Constraints: a fixed budget that allows only open-source tools and the college's existing hardware.

4. Work Breakdown Structure and Timeline

Phase	Key Activities	Duration	Deliverable
1. Requirements	Interviews with librarian and students, use case identification, data cleaning	2 weeks	Software Requirement Specification
2. Design	Database schema, UI wireframes, system architecture	2 weeks	Design document
3. Development	Catalogue, membership, circulation and fine modules	4 weeks	Working modules
4. Testing	Unit, integration and user acceptance testing	2 weeks	Test report

Phase	Key Activities	Duration	Deliverable
5. Deployment	Installation, data migration, staff training	1 week	Live system
6. Maintenance	Bug fixes and minor enhancements after go-live	Ongoing	Maintenance log

The total planned duration for initial delivery is about eleven weeks. A Gantt chart drawn from this table would show requirements and design partly overlapping, and testing starting for each module as soon as its development is complete. Milestones are placed at the end of each phase, with a formal review meeting with the librarian at each milestone.

5. Team Roles and Responsibilities

Role	Responsibility	Count
Project Manager	Planning, schedule tracking, stakeholder communication, risk log	1
Business Analyst	Requirement gathering, SRS preparation, change control	1
Designer	UI wireframes and usability review	1
Developers	Front-end, back-end and database implementation	2
Tester	Test case design, execution and defect tracking	1
Librarian (client)	Acceptance testing and final sign-off	1

6. Risk Management

Risk	Likelihood	Impact	Mitigation
Incomplete or inconsistent book data	High	High	Data cleaning in phase 1; validation rules on entry
Delayed user feedback	Medium	Medium	Fixed review meetings at the end of each phase
Low adoption by staff	Medium	High	Hands-on training and a simple, consistent interface
Server downtime during exam period	Low	High	Scheduled backups and a documented recovery procedure

7. Quality and Communication Plan

- Weekly status reports from the Project Manager to the Head of Department.
- Change requests recorded in a change log and approved by the librarian before implementation.

- Coding standards and version control are used by developers to keep the codebase consistent.
- Each deliverable has a review checklist to confirm completeness before the next phase begins.

8. Result and Conclusion

A complete project plan has been prepared with defined scope, phased timeline, team roles, risk controls and a communication schedule. The plan provides a baseline against which progress can be measured, and it gives every stakeholder a clear view of responsibilities and deliverables. Careful scoping, particularly the decision to exclude e-books and payments, is the main reason the eleven-week schedule is realistic.